Uniform mesh refinement underestimates the frequencies of the DeviceLayout.jl transmon: an independent finite-element cross-check and re-optimization

Robb Walters
Independent Researcher

October 2026

Abstract

The frequencies that the SingleTransmon example of DeviceLayout.jl publishes for an initial and an optimized design are discretization-limited. We reproduce them with Palace within 0.7% and cross-check Palace with geode-fem, an independent solver; on shared meshes the two agree to 2×10^{-4}% on the qubit and 0.03% on the resonator. All frequencies are linear eigenmodes. Adaptive Palace runs, still rising, put the qubits 7.0% and 8.0% above their published values without extrapolation. On the initial design, a two-sided island-capacitance bound for the grounded box, read through a circuit model, caps the qubit at 4.477 GHz (+8.1%). Extrapolated at order 1 in mesh size, the fastest rate the sheet edges allow on uniform meshes, two refinement families put the qubits 7.3–9.6% and the resonators 2.7–3.3% higher; these estimates lean low and rest on Palace's levels alone, since geode-fem ran only order 1 on the two coarsest meshes. At these frequencies the 3.0/4.0 GHz design needs a capacitor 19–20% longer than published (17–39% within the band). Counting every Palace run, probes included, Newton's method meets DeviceLayout's criterion in 3 runs, with geode-fem's shape derivative or a geode-fem-calibrated scaling law in its Jacobian, and in 4 without geode-fem; PRIMA takes 10 to 12 in our environment.

1 Introduction

Superconducting-qubit layouts are designed by iterating field solves against frequency targets, a loop that QDesignOptimizer (Eriksson et al., 2025) steers with analytic models. DeviceLayout.jl (AWS Center for Quantum Computing, 2025a) builds the layout and its mesh, and Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based solver (Anderson et al., 2021), computes the eigenmodes. DeviceLayout's SingleTransmon example (AWS Center for Quantum Computing, 2025b; Peairs and Carson, 2025) runs the loop end to end. It publishes an initial transmon-plus-resonator design at 4.14 and 5.591 GHz and a design optimized for targets of 3.0 and 4.0 GHz, at 3.005 and 3.982 GHz, found with a derivative-free optimizer in nine Palace runs. Like every frequency in this paper, these are eigenfrequencies of a linear model with the junction as a lumped inductor: the qubit frequency is the junction's linearized (plasma) mode, not the transmon's 0–1 transition f_{01}, which lies below it by about the charging energy E_C/h (Section 3.3).

Those frequencies come from one discretization, and the example's authors say what that implies. They run Palace at second order "to reduce sensitivity to the discretization" and note that this "does not mean that the eigenfrequencies found by Palace will have converged to their true values" (AWS Center for Quantum Computing, 2025b). This paper measures how far from converged the published frequencies are. Our main result is that the published frequencies are discretization-limited. Palace's adaptive refinement reaches qubits 7.0% and 8.0% above the two published values without

extrapolation, and its uniform ladders, in two refinement families, extrapolate to qubits 7.3–9.6% above them; every uniform level we could afford lies below those estimates. geode-fem reproduces every level of those ladders it can reach, the two coarsest meshes at order 1, to 10^{-4}% on the qubit, and a two-sided bound on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model.

The second code is GEODE-FEM ("geode-fem"), a Rust finite-element solver on the Burn tensor framework (Tracel AI, 2026). It solves the same $H(\mathrm{curl})$ eigenproblem as Palace with a separate implementation, and it computes design derivatives, among them the shape derivative of an eigenfrequency. Given the same input mesh, it reproduces Palace's modes (Section 4), and on the $h/2$ meshes of Section 5 the two codes solve the same discrete problem apart from the junction term. This gives an independent same-mesh cross-check of the Palace/DeviceLayout.jl transmon stack, complementing the earlier checks of Palace by SQDMetal (Sommers et al., 2025) and Ye et al. (2025).

Our contributions are:

- **A refinement study of a published design example**, which puts its qubits 7.3–9.6% above their published values, with a second refinement family and adaptive runs on both designs checking the qubit estimates; the finer levels and the extrapolations are Palace's (Section 5).
- **Two-sided capacitance bounds with the sapphire tensor.** An equilibrated-flux lower bound and conforming upper bounds bracket the island capacitance C_{isl} of the unmodified device, without the junction capacitance, in $[79.56, 79.77]$ fF with the vacuum box grounded, as in the eigenproblem, and in $[79.56, 79.76]$ fF with it natural; under a two-node circuit model the lower bound caps the initial qubit at 4.477 GHz (Section 6).
- **A re-optimized design, with controls.** At the extrapolated frequencies the 3.0/4.0 GHz design needs $c \approx 1451$–$1466\,\mu$m against the published $1217.678\,\mu$m. Newton's method, with and without geode-fem input, Broyden's method and PRIMA are compared by the Palace runs they need in one environment (Section 7).
- **Same-input-mesh agreement with Palace**, with the 17.49 GHz mode that both codes share traced to the isotropic sheet term of the port model (Section 4).

2 Related Work

Cross-checks of Palace. SQDMetal (Sommers et al., 2025) cross-checks a Palace workflow against COMSOL and Ansys with mesh-convergence studies, and Ye et al. (2025) compare a Palace workflow with cryogenic measurement, matching resonator frequencies to within 0.3% with a mesh refined at the CPW gaps and fourth-order elements, a setting their convergence study puts within ±0.3% of extrapolated values; energy-participation analyses likewise check convergence over adaptive passes (Minev et al., 2021a). Both cross-checks use models built independently for each tool. Our comparison instead holds the input mesh and the junction model fixed and changes the solver implementation, and it adds a refinement study of a published design: code-to-code comparison and solution verification (Oberkampf and Roy, 2010), not validation against measurement. As of October 2026, neither Palace nor DeviceLayout.jl has a methods publication; we cite their repositories and documentation.

Design sensitivities and qubit-design optimization. Adjoint sensitivity analysis of frequency-domain electromagnetic solvers is an established field (Nikolova et al., 2004), including shape

adjoints for Maxwell cavity eigenvalues (Akçelik et al., 2005); Palace exposes no derivatives. In qubit design, SQcircuit (Rajabzadeh et al., 2023) and SuperGrad (Wang et al., 2025) differentiate circuit Hamiltonians, SQuADDS (Shanto et al., 2024) looks designs up in what its authors call a "validated design database", and Eun et al. (2023) shape-optimize transmon pads with FEM and a derivative-free optimizer. DeviceLayout's example optimizes with PRIMA (Zhang, 2023), a reference implementation of Powell's model-based derivative-free methods; with bounds only it runs BOBYQA (Powell, 2009).

Two-sided capacitance bounds and circuit models. Bracketing a capacitance between complementary energy principles is classical (Specogna, 2011). Our lower bound uses the hypercircle idea of Prager and Synge (1947) with Raviart–Thomas fluxes (Raviart and Thomas, 1977) equilibrated patch by patch (Braess and Schöberl, 2007; Ern and Vohralík, 2015). The method is not new; we apply it to an island capacitance with a floating conductor on a production qubit layout. A web search in October 2026 of the complementary-bound and qubit-design literature (queries such as "guaranteed capacitance bounds qubit") found no two-sided capacitance bound in the qubit-design literature. The two-node circuit of Section 3.3 is a minimal instance of the quasi-lumped route (Minev et al., 2021b), which partitions a device into lumped or quasi-distributed cells, simulates each and reduces the composite to coupled subsystems; the energy-participation route (Minev et al., 2021a) instead quantizes the distributed eigenmodes.

3 Solvers, Device and Models

3.1 geode-fem

geode-fem assembles first-order Nédélec (Nédélec, 1980) and scalar P1/P2 element operators as batched $[n_{\text{elem}}, k, k]$ tensor operations on Burn. Its eigensolver is shift-invert Lanczos (Lehoucq et al., 1998) on a real symmetric pencil, factorized with faer (Kazdadi, 2026). The two codes share the formulation, the lumped junction port, the materials and the input mesh, and geode-fem follows a specification of Palace's operators (Section 8); assembly, the eigensolver and the linear algebra are separate implementations.

The eigenfrequency derivative used in Section 7 needs no adjoint solve, because the pencil is symmetric: for a simple eigenpair (λ, x) with $x^{\mathsf{T}} M x = 1$,

$$\frac{d\lambda}{d\theta} \;=\; x^{\mathsf{T}}\Big(\frac{\partial K}{\partial \theta} - \lambda\,\frac{\partial M}{\partial \theta}\Big)x, \tag{1}$$

the classical eigenvalue derivative of a symmetric generalized eigenproblem (Fox and Kapoor, 1968). Here K and M include the junction terms of Eq. (2); the junction does not move under the capacitor-length motion of Section 7, so those terms do not contribute. geode-fem evaluates $\partial K/\partial\theta$ and $\partial M/\partial\theta$ along a node velocity $\partial X/\partial\theta$ with a hand-written forward-mode dual-number twin of its Nédélec kernel (tested against it to 10^{-10}); Burn's autodiff tape is not used.

3.2 Device, mesh and materials

The device is DeviceLayout.jl v1.15.0's SingleTransmon example (AWS Center for Quantum Computing, 2025a): a transmon (Koch et al., 2007) with a meander readout resonator and a feedline on sapphire, in a vacuum box (Figure 1). Its two design parameters are the transmon's `capacitor_length` c and the resonator's `total_length` L; the published initial design has $c = 620\,\mu$m and $L = 5000\,\mu$m. The sapphire permittivity is $\mathrm{diag}(9.3, 9.3, 11.5)$ with the optic axis normal to the chip; the configuration rotates the material axes about that normal, which leaves a uniaxial tensor unchanged, so the lab tensor is diagonal. Where a scalar model is used it is the trace average $\bar{\varepsilon} = 10.03$.

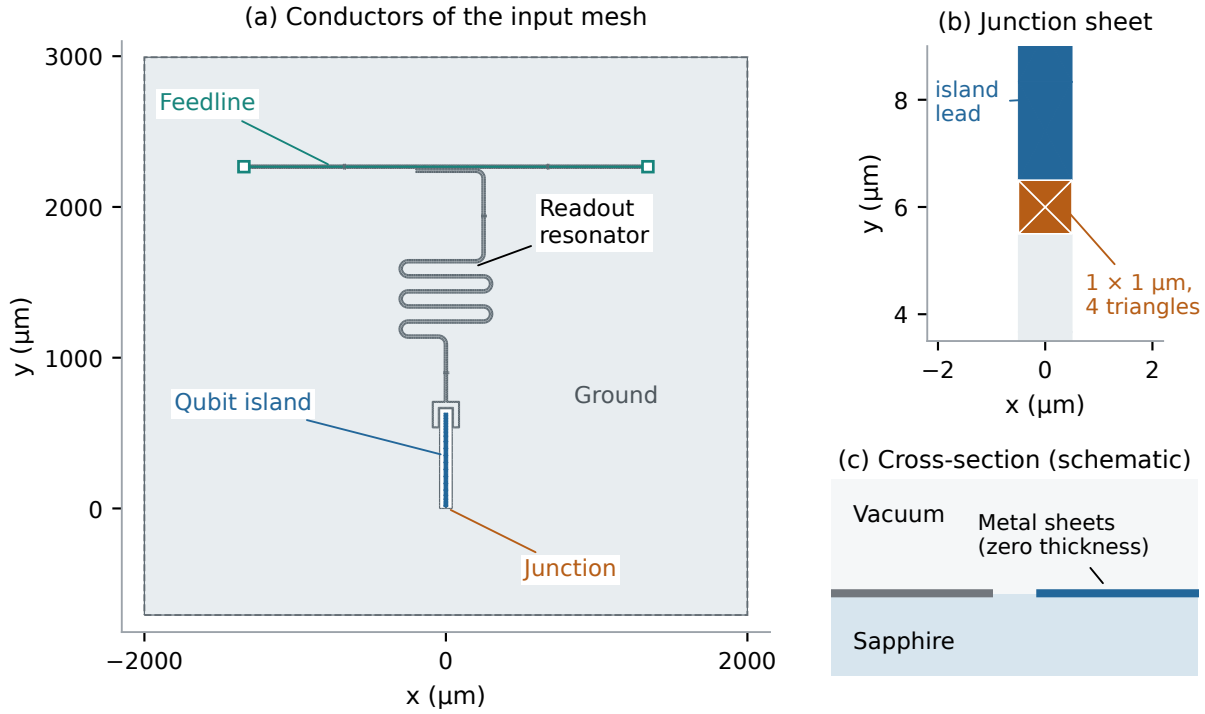


Figure 1: Benchmark geometry, drawn from the SHA-256-pinned MSH 4.1 input mesh of the DeviceLayout.jl v1.15.0 `SingleTransmon` device. (a) Top view of the three metal conductors of Section 6: the feedline (teal), the qubit island (blue), and the ground plane with the meander resonator (gray); squares mark the two readout ports. (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its four-triangle triangulation. (c) Material cross-section, schematic and not to scale. Both solvers read this file (22,684 nodes; 133,314 tetrahedra).

Two kinds of mesh appear. The *fixture* is a first-order mesh of the initial geometry written by an earlier DeviceLayout.jl v1.15.0 driver, `reference/julia/generate_transmon_fixture.jl` (22,684 nodes, 133,314 tetrahedra), pinned by SHA-256; Section 4 and the capacitance brackets of Section 6 use it and its refinements, which leave its polygonal surfaces unchanged. For the cross-check and the re-optimization, DeviceLayout meshes each geometry with the two parameters exposed, at first order for the ladders and at its own second order for the runs that reproduce its discretization; curving the faces changes the qubit by at most 0.0001% and the resonator by about 0.05%. On the initial geometry the fixture and DeviceLayout's mesh give the same Palace qubit at h (3.4527 against 3.4520 GHz, 0.02%). PEC is applied on the metal sheets and the box; the readout ports are left open in every eigenproblem except the DeviceLayout-configuration runs of Section 5.

3.3 Junction and circuit models

The Josephson junction is a linear lumped element, $L_J = 14.860\,\text{nH}$ in parallel with $C_J = 5.5\,\text{fF}$, on the four-triangle sheet Γ of Figure 1b, as in DeviceLayout's Palace configuration. With port length ℓ and width w the uniform lumped-port construction adds

$$K_{\text{port}} = \frac{\ell}{w\,L_J}\,S_\Gamma \qquad \text{and} \qquad M_{\text{port}} = \frac{C_J\,\ell}{w}\,S_\Gamma \tag{2}$$

to the pencil $(K + K_{\text{port}})x = \omega^2(M + M_{\text{port}})x$, where S_Γ is the isotropic surface matrix on Γ. Palace's `LumpedPort` adds the same terms with the same scalings. geode-fem's rank-1 port-voltage shunt replaces S_Γ by $b\,b^{\mathsf{T}}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\, dS$ the port-voltage functional; on the fixture it moves the qubit by 6×10^{-5} %, keeps the other physical modes to four decimals, removes the artifact mode of Section 4, and is the junction model of every geode-fem eigensolve in Sections 5 and 7.

The electrostatic model splits the metal into three conductors: ground with the resonator, the floating feedline, and the island. Its island capacitance with the feedline floating and neutral, $C_{\text{isl}} = C_{ii} - C_{if}^2/C_{ff}$, excludes the junction. In the convention of Koch et al. (2007) the transmon's total capacitance is $C_\Sigma = C_{\text{isl}} + C_J$, and it sets the charging energy $E_C = e^2/2C_\Sigma$. A two-node circuit connects it to the eigenproblem: a bare qubit $1/(2\pi\sqrt{(L_J + L_s)C_\Sigma})$, with L_s the series inductance of the junction's current path, coupled with coefficient k to the resonator. At $L_s = k = 0$ this is the plasma frequency $\sqrt{8E_JE_C}/h$, and the transmon's $f_{01} \approx \sqrt{8E_JE_C}/h - E_C/h$ lies below it. On the fixture L_s and k were fitted on three and two discretizations; extrapolated, they give bands of 0.113–0.125 nH and 0.0227–0.0310, which are estimates.

4 Same-Input-Mesh Agreement With Palace

Both codes read the fixture with the same materials and junction model at first order (Table 1). Palace's configuration schema defines `Target` as the frequency above which its eigensolver searches (Palace developers, AWS Center for Quantum Computing, 2023), so the six modes above 4.5 GHz come from a run with `Target` = 4.5 GHz and the qubit from one with `Target` = 2 GHz. All six agree to $\leq 0.033\%$, and the qubit mode to 5×10^{-5} %, with port participation 0.9942 in geode-fem and port EPR 0.99424 in Palace, the energy-participation ratio of the junction, the fraction of the mode's inductive energy stored in it (Minev et al., 2021a). Both codes use the isotropic sheet term of Eq. (2) here.

The two codes read the same input mesh but do not solve the same unknowns. Palace cracks each metal sheet, but not the junction sheet, into two sides by duplicating its vertices. Where an edge inside a sheet joins two outline vertices, which are not duplicated, the sides cannot be separated, and Palace bisects the elements on both sides of that edge, in two passes on the fixture ("local bisection for under-resolved interior boundaries" in its log; `palace/utils/geodata.cpp`). Such edges lie where a sheet is one element wide, almost all on the resonator and feedline sheets (Table 1), so the added tetrahedra are concentrated there, though some touch the island. Cracking alone leaves the PEC problem unchanged: each volume edge attaches to one copy of a duplicated vertex, so the edges that cracking adds are copies of edges in a sheet, and the PEC condition eliminates them. Only the bisection changes the unknowns.

The 17.49 GHz mode is a shared artifact of the isotropic sheet term: the junction patch has one free node whose hat-function gradient carries no port voltage but stores energy in S_Γ, with Rayleigh quotient $1/(L_JC_J)$, or $f_{LC} = 17.60$ GHz, lowered by the adjacent volume mass; Palace gives it a port EPR of 2.49×10^{-8}. This mechanism is our inference, supported by a synthetic test; the rank-1 shunt removes the mode. The fixture is coarse: Palace's own refinements of it move the qubit from 3.4527 to 3.9528 GHz at $h/2$ (+14.5%) and to 4.1721 GHz at second order on h (+20.8%), and the resonator by +6.1% at $h/2$, so the agreement above is a code-to-code comparison on one discretization.

5 The DeviceLayout Cross-Check

The published designs. The DeviceLayout.jl blog (Peairs and Carson, 2025) reports the initial design ($c = 620\,\mu$m, $L = 5000\,\mu$m) at $[4.14, 5.591]$ GHz and the final design ($c = 1217.678\,\mu$m,

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes, 133,314 tetrahedra), first order in both solvers, junction as the lumped $L_J \parallel C_J$ shunt. Counted the same way, as all first-order Nédélec unknowns (edges) including those on PEC sheets, geode-fem's mesh has 156,863 (133,108 interior) and Palace's has 183,891, after it cracks the metal sheets and adds 4,913 elements on them by bisection, triggered by 1,007 edges, 993 of them on the resonator's ground plane and the feedline. Both codes are printed to geode-fem's archived three decimals (six for the qubit); Δ is the archived relative difference, computed from Palace's full-precision value. Sources: `benchmarks/transmon_eigen/results.toml` and `benchmarks/transmon_eigen/palace_crack/`.

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
qubit	transmon qubit	3.452751	3.452749	0.00005
resonator	readout resonator	5.153	5.151	0.032
mode 2	box/cavity	15.465	15.461	0.029
port sheet	port-sheet artifact	17.490	17.490	0.001
mode 4	box/cavity	18.693	18.692	0.007
mode 5	box/cavity	20.703	20.698	0.026
mode 6	box/cavity	26.088	26.081	0.027

$L = 7243.955\,\mu\text{m}$) at $[3.005, 3.982]$ GHz, reached by PRIMA in nine Palace runs "within 1% of our target frequencies". The example source (AWS Center for Quantum Computing, 2025b) sets the targets to $[3.0, 4.0]$ GHz, works in the variables $x_1 = \sqrt{620\,\mu\text{m}/c}$ and $x_2 = 5000\,\mu\text{m}/L$, and stops when $\text{err} = \frac{1}{2}\sum_m (f_m - t_m)^2/t_m^2 < 10^{-4}$. Its frequencies are computed with Palace on DeviceLayout's mesh at mesh order 2 and solver order 2, with a first-order absorbing far field and two $50\,\Omega$ readout ports.

Reproduction at DeviceLayout's discretization. Palace on DeviceLayout's second-order mesh with the example's configuration file gives $[4.1683, 5.6132]$ GHz for the initial design and $[3.0164, 3.9956]$ GHz for the final one, 0.68% and 0.40% above the published initial pair and 0.38% and 0.34% above the final pair. The same cells with the PEC-box model used everywhere else differ from the configuration-file runs by at most 0.003%. Neither the blog nor the example records the Palace version, mesher build or platform of the published runs (ours are in Section 11). Every published value lies below ours, as a slightly coarser published mesh would make it, but which difference causes the 0.34–0.68% gap is not diagnosed.

The ladder and its rate. For each geometry, Palace runs at order 1 on DeviceLayout's first-order mesh h and its uniform refinements $h/2$ and $h/4$ (gmsh's eight-way split, up to 12.6 million unknowns), and at order 2 on h and $h/2$; geode-fem runs at order 1 on h and $h/2$, on the same meshes. Every computed series rises under refinement (Figure 2); this is an observation, not a guarantee. At the edge of a zero-thickness metal sheet the field behaves as $E \sim r^{-1/2}$ (Meixner, 1972), so near the edge the eigenfunctions lie in H^s only for $s < 1/2$, and on uniform meshes the eigenvalue error decays as h^{2s} at any element order (Boffi, 2010), slower than $O(h)$. The order-1 ladders show pre-asymptotic orders of 0.61–0.62 (qubit) and 0.70–0.73 (resonator).

Extrapolation rule. The *estimate* is the mean of two extrapolations at order 1 in the frequency, $2f_2 - f_1$: one from the order-1 pair ($h/2$, $h/4$) and one from the order-2 pair (h, $h/2$). Order 1 is the fastest rate the theory allows on uniform meshes, so the rule makes the smallest correction consistent with it, and its estimates lean low; for the order-2 pair the order is taken from theory, because two levels give no observed order. The *band* spans six values, the same two pairs taken three ways (values for the initial qubit):

- at order 1 in the frequency, the two values of the estimate (4.432 and 4.455 GHz);
- at order 1 in the eigenvalue variable $\lambda \propto f^2$, $\sqrt{2f_2^2 - f_1^2}$, which lie lower for a series that rises under refinement (4.415 and 4.451 GHz); the order-1 value sets the low end, except on family-b resonators, where the order-2 value does;
- at the observed order of the order-1 ladder's three levels (0.62 and 0.61 for the family-a qubits), borrowed by the order-2 pair on the assumption that both element orders share the pre-asymptotic rate, which lie higher (4.677 and 4.582 GHz); the order-1 value sets the high end.

The band is an extrapolation range, not a bound. For the initial qubit, the circuit-model cap of Section 6 trims it at 4.477 GHz.

The rule is Richardson extrapolation (Richardson, 1911) at the asymptotic order rather than the observed one. Roache's grid convergence index (Roache, 1994) scales the correction at the observed order by a safety factor: on the initial qubit's order-1 ladder, the fine-grid index with a factor of 1.25 is centred on 4.155 GHz with a half-width of 0.65 GHz, which reaches 4.81 GHz, above the circuit-model cap. The rule was fixed after the static route had been computed; extrapolation at the observed order alone puts the initial qubit at 4.63 GHz, above the cap, and the observed-order values are kept only as the band's high side. The agreement of the two routes is therefore a consistency check, not a blind prediction.

Table 2: Published frequencies, Palace at DeviceLayout's discretization (DL), the finest level of Palace's adaptive run (a measured value, still rising), and the extrapolated estimates with their bands, in GHz, for the two refinement families of DeviceLayout's mesh h (a: gmsh's eight-way split; b: Palace's uniform refinement). Percentages are relative to the published value. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` (family a) and `benchmarks/transmon_devicelayout_controls/results.toml` (adaptive, family b).

			Adaptive,	Extrapolated estimate [band]	
	Published	DL	finest level	Family a	Family b
Initial qubit	4.14	4.1683	4.4297 (+7.0%)	4.444 (+7.3%) [4.415, 4.677]*	4.472 (+8.0%) [4.458, 4.510]
Initial resonator	5.591	5.6132	5.6986 (+1.9%)	5.740 (+2.7%) [5.723, 5.817]	5.776 (+3.3%) [5.768, 5.800]
Final qubit	3.005	3.0164	3.2468 (+8.0%)	3.265 (+8.7%) [3.236, 3.471]	3.292 (+9.6%) [3.278, 3.331]
Final resonator	3.982	3.9956	4.0552 (+1.8%)	4.088 (+2.7%) [4.077, 4.139]	4.113 (+3.3%) [4.106, 4.125]

*Under the circuit model of Section 6, the cap of 4.477 GHz trims the band to [4.415, 4.477].

Result. The initial qubit extrapolates to 4.444 GHz and the final qubit to 3.265 GHz, 7.3% and 8.7% above the published values; both resonators extrapolate to 2.7% above theirs (Table 2). Every published value lies below its band. Measured against the estimates of the two families, DeviceLayout's discretization sits 6.2–6.8% and 7.6–8.4% low on the two qubits and 2.2–2.8% low on the resonators. With no extrapolation, the highest uniform levels, order 2 on $h/2$ (7.0 million unknowns on the initial geometry; order 1 on $h/4$ has more unknowns and lower values), already lie 4.2% and 4.7% above the published qubit values. The estimates rest on Palace's order-1 $h/4$ and order-2 levels, which geode-fem did not run. On the levels it did run, geode-fem and Palace agree on the qubit to 10^{-4} % on both geometries. On the resonator they differ by 0.03% at h, where Palace adds 4,946 (initial) and 5,944 (final) elements by local bisection, and not at all in the six archived decimals at $h/2$, where Palace's logs report no bisection and the two codes differ only in the junction term (Section 3.3; the counts are in the logs). Three checks test the estimates, and all three concern the qubit.

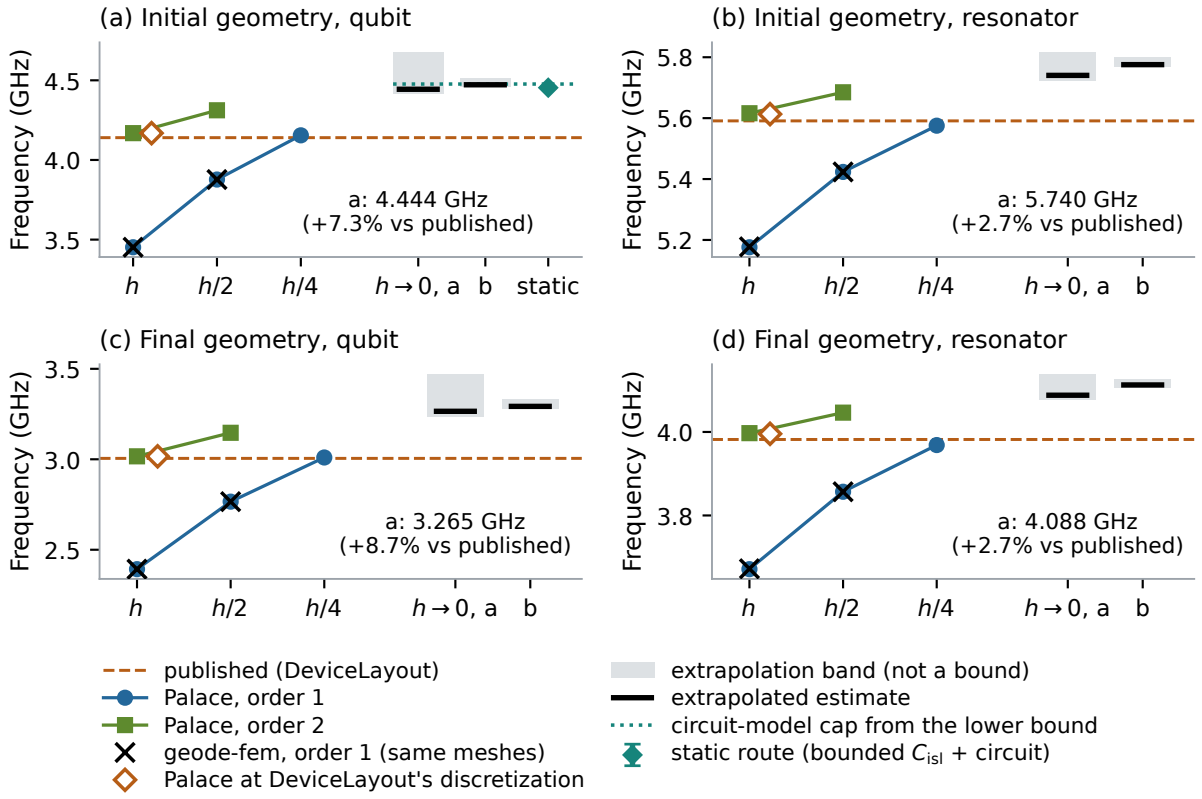


Figure 2: Resolution ladders of the two published DeviceLayout.jl SingleTransmon designs: Palace at orders 1 and 2 on DeviceLayout's mesh h and its uniform refinements, and geode-fem at order 1 on the same meshes (crosses); the open diamond, just right of h, is Palace at DeviceLayout's own discretization, that of the published values (dashed). The "$h \to 0$" slots are categorical: the estimate with its band for the gmsh refinement family (a, labelled) and for Palace's refinement (b). In (a), the static route of Section 6 (grounded box) and the circuit-model cap, which trims the untrimmed bands drawn. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

A second refinement family. Family b refines the same h mesh with Palace's own uniform refinement instead of gmsh's split (family a). It first bisects where a sheet is one element wide (Section 4), so it is graded slightly toward the sheets, a plausible reason for its higher observed orders, 0.91 (initial) and 0.90 (final) against family a's 0.62 and 0.61. Its $h/2$ qubit is 3.9469 GHz against gmsh's 3.8771 GHz, and on the fixture the two refinements give 3.9528 and 3.8939 GHz, so the refinement scheme accounts for 1.5–1.8% of the $h/2$ level and the choice of h mesh for only 0.15–0.43%. Its qubit estimates lie 0.64% and 0.83% above family a's, and every family-b band lies inside the family-a band (Table 2). We therefore quote the initial qubit as 4.444–4.472 GHz, 7.3–8.0% above the published value, and the final qubit as 3.265–3.292 GHz, +8.7% to +9.6% over the published 3.005 GHz.

Adaptive refinement. A further check comes from Palace's adaptive mesh refinement, run on each geometry from the same h mesh at order 2, with the error indicator summed over the qubit and resonator modes, Dörfler marking at 0.7 and nonconformal refinement (Figure 3 for the initial geometry). On the initial geometry, in five solves from 0.95 to 11.0 million unknowns, the qubit rises from 4.1683 to 4.4297 GHz, 0.32% below the estimate and inside the band; the third solve, at

2.2 million unknowns, already exceeds the highest uniform level (4.3118 GHz at 7.0 million). On the final geometry, from 1.17 to 14.6 million unknowns, the qubit rises from 3.0164 to 3.2468 GHz, 8.0% above the published 3.005 GHz, and the third solve again exceeds the highest uniform level (3.1884 against 3.1464 GHz). The finest final level lies inside the band, 0.56% and 1.39% below the two family estimates. Both finest levels are measured values, not extrapolations, and neither run reached a plateau: each stopped when its last solve passed the run's size limit of 10^7 unknowns, not on its tolerance, and the last step still moved the qubit by +0.55% (initial) and +0.63% (final). A three-point fit $f_\infty - CN^{-\alpha}$ through the last three solves gives 4.4552 GHz ($\alpha = 0.80$) and 3.2672 GHz ($\alpha = 0.83$), but +0.1% on the finest level moves them by 24 and 16 MHz, so they are ill-conditioned estimates. The adaptive resonators rise more slowly and still lie below their bands, so the adaptive runs check the qubit estimates only.

A common-limit fit. The last check fits the order-1 ($h/2$, $h/4$) and order-2 (h, $h/2$) series of each ladder with one common limit and order. This fit is exactly determined; it lies 0.5–1.3% above the rule's estimate on every first-family series, as expected of a rule that leans low, and +0.1% on the order-2 $h/2$ level alone moves the initial-qubit fit by 20 MHz.

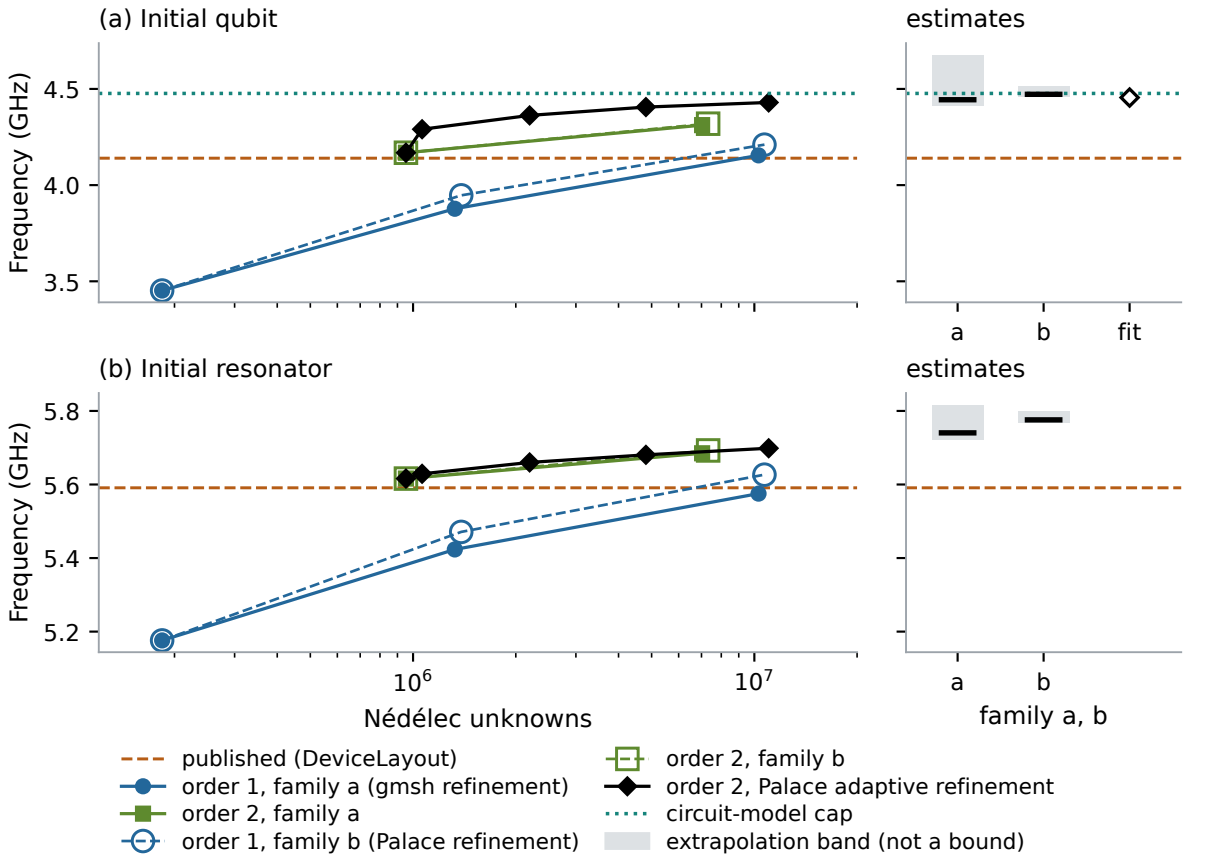


Figure 3: Checks of the extrapolated estimate on the initial geometry. Left: every Palace level against the number of Nédélec unknowns, for the two refinement families of h (filled: gmsh's split, as in Figure 2; open: Palace's) and Palace adaptive refinement from h at order 2 (diamonds). Right: the estimates of the two families with their bands and, for the qubit only, the adaptive run's ill-conditioned fit (the resonator fit is unreliable). In (a), the circuit-model cap. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

6 A Static Route: A Bounded Island Capacitance

The bound. For the reduced electrostatic problem (island at potential 1, ground and resonator at 0, feedline floating with zero net charge, and zero normal flux on the vacuum box), the potential ϕ satisfies $0 \le \phi \le 1$ by the maximum principle, and $C_{\mathrm{isl}} = \int \varepsilon \nabla\phi \cdot \nabla\phi$ with $\varepsilon = \varepsilon_0\varepsilon_r$. Every conforming finite-element energy is an upper bound on C_{isl}, because C_{isl} is the minimum of the energy over admissible potentials. For the lower bound, take any piecewise polynomial field D on the mesh cut along the zero-thickness sheets. Integrating by parts and using $0 \le \phi \le 1$ on every term gives

$$\int D \cdot \nabla\phi \;\ge\; N(D) \;=\; Q_{\mathrm{island}}(D) - |Q_{\mathrm{feed}}(D)| - \|\operatorname{div} D\|_{L^1} - \|[D\cdot n]\|_{L^1(\mathcal{F})} - \|D\cdot n\|_{L^1(\mathcal{B})}, \quad (3)$$

where Q are the fluxes of D into the two conductors (for the exact field $D = \varepsilon\nabla\phi$, the negative of the physical displacement, they are the conductor charges), $[D\cdot n]$ is the normal jump on the interior faces $\mathcal{F}$ off the sheets, and $D\cdot n$ on the box faces $\mathcal{B}$ is the one-sided flux, which the zero-flux condition requires of the exact field but not of D. Cauchy–Schwarz, $\left(\int D\cdot\nabla\phi\right)^2 \le \int D\cdot\varepsilon^{-1}D \int \varepsilon\nabla\phi\cdot\nabla\phi$, then gives, whenever $N(D) > 0$,

$$C_{\mathrm{isl}} \;\ge\; \frac{N(D)^2}{\int D\cdot\varepsilon^{-1}D}. \quad (4)$$

The bound holds for any D, so solver tolerance can only weaken it. Each L^1 norm is bounded above element by element and face by face, $\|p\|_{L^1(K)} \le |K|^{1/2}\|p\|_{L^2(K)}$, with $\|p\|_{L^2}^2$ integrated exactly by quadrature. We take D from the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three, with an independent Crouzeix–Raviart flux as a check. The two bounds hold by theorem, for the exact arithmetic of the discretization on the polygonal geometry of the mesh, with lower bounds floored and upper bounds ceiled; the implementation is checked by tests, not formally verified. The tests compare it with an exact parallel plate and an analytic coaxial bracket, and mutation tests reject a grounded-feedline or jump-free lower bound. The bound is evaluated in plain f64, not interval arithmetic; the worst-case summation error is about 10^{-10} relative, far below the rounding. Refinement does not move the polygonal surfaces, so all meshes of one geometry bound the same exact capacitance, and a family bracket is the largest lower bound and the smallest upper bound.

Two box models. The bound above leaves the vacuum box natural; the eigenproblem makes it PEC (Section 3.2). Constraining the potential on the box, to zero or to a floating constant, shrinks the set of admissible potentials, so it can only raise C_{isl}: the natural-box lower bound, and the cap below that uses only it, hold for the PEC box too. For an upper bound we re-solve the conforming P2 problems on the same meshes with the box grounded.

The unmodified device. Uniform refinement of the fixture with P1 and P2 elements gives the upper bounds of Figure 4: with the scalar model, from 137.706752 fF at P1 on h to 83.015285 fF at P2 on $h/2$, and with the tensor from 136.684731 to 85.008425 fF. On adaptively refined meshes the lower bound closes the gap: the island capacitance of the unmodified initial design is bounded by $[77.40, 77.61]$ fF with the scalar $\bar{\varepsilon}$ and by $[79.56, 79.76]$ fF with the sapphire tensor (Figure 4). The bounded tensor-minus-scalar difference, computed from the unrounded bounds, is 1.95–2.35 fF, or 2.5–3.1% of the scalar value. With the box grounded, the tensor C_{isl} is bounded by $[79.56, 79.77]$ fF. On the finest mesh, grounding the box raises the conforming value by 0.0065 fF, 8×10^{-5} relative; that is an estimate of the effect of the box model, whose size is bounded only by $[0, 0.21]$ fF; the same bracket

covers a floating box, which lies between the two. With C_J added, the transmon's $C_\Sigma = C_{\mathrm{isl}} + C_J$ is bounded by $[85.06, 85.27]$ fF for the eigenproblem's grounded box (tensor; $[85.06, 85.26]$ fF natural). The fixture itself overstates the tensor C_{isl} by at least 56.92 fF, which is at least 71.3% of the exact value, and the scalar one by at least 77.4%.

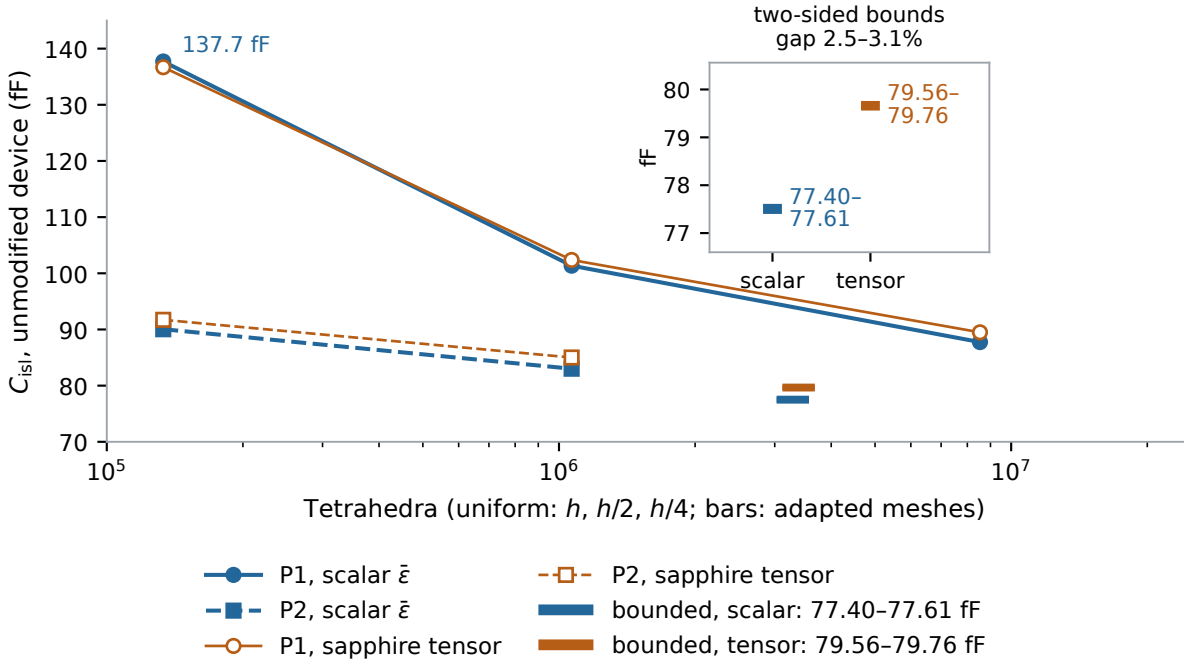


Figure 4: Mesh convergence of the island capacitance C_{isl} of the unmodified device under uniform refinement of the fixture, with the scalar trace-averaged permittivity (blue) and the sapphire tensor (orange). Every marker is a conforming upper bound on the exact capacitance of the meshed polygonal geometry. The short bars are the two-sided bounds from adapted meshes, each drawn at the tetrahedron count of the mesh that sets its lower bound and magnified in the inset; the inset title gives the bounded tensor-minus-scalar gap. All values are for the natural box. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml`.

From capacitance to the qubit frequency. Reading the grounded-box tensor bound through the two-node circuit of Section 3.3, over every corner of the C_{isl}, L_s and k bands, with the resonator at its extrapolated estimate, gives the initial dressed qubit in $[4.449, 4.458]$ GHz; the natural-box bound moves its low end by 0.3 MHz. That range lies inside the eigenmode band of Section 5 and inside the range of the two family estimates, and the first family's estimate is 0.22% below its midpoint, a consistency check for the reason given in Section 5. The range is an estimate, because L_s, k and the resonator estimate are.

A model-dependent cap. Under the circuit model, a series inductance and a coupling to a higher mode can only lower the qubit, so the lower bound $C_{\mathrm{isl}} \geq 79.56$ fF, that is $C_\Sigma \geq 85.06$ fF, caps it at $1/(2\pi\sqrt{L_J C_\Sigma}) = 4.477$ GHz. The cap is rigorous only given that model (one island node with $L_J \parallel C_J$, $L_s \geq 0$, coupling only to modes above the qubit, a quasi-static C_Σ) and rests on the fixture, not on the ladder meshes, which agree with it to 0.02% at h, so it is model-dependent and not in the Bounded tier. It does not depend on the extrapolation rule. Every computed level and both family estimates lie below it; the first family's common-limit fit, 4.4802 GHz, exceeds it

by 3.6 MHz, within that fit's conditioning. With no extrapolation, the adaptive level and the cap put the initial qubit between 4.4297 and 4.477 GHz, +7.0% to +8.1% over the published value; the lower end is a measured level, the upper end holds only under the circuit model.

Why the qubit is approached from below. The qubit mode is quasi-static, and the quasi-static part of a first-order Nédélec field is the gradient of a P1 potential, which can only overstate the capacitance, so the computed qubit should approach its limit from below: on the fixture the P1 bound (136.68 fF, with C_J added) overstates C_Σ by a factor of 1.67, and the family estimates exceed the fixture's order-1 qubit by 1.66–1.68 in f^2. This is a heuristic, not a proof: it omits L_s, the coupling, the rotational field and the distributed resonator.

7 Re-Optimization and Its Controls

At the extrapolated frequencies the published final design misses its targets. We repeat DeviceLayout's optimization against its own discretization, to compare optimizers in one environment, and against the extrapolated frequencies, to find the design that meets the targets.

The capacitor_length derivative. DeviceLayout builds the island upward from the fixed junction, so a change of c translates every node above the island and stretches the island band. geode-fem turns this motion into a node velocity $\partial X/\partial c$, checks that every metal outline node in a stretched band moves along its own edge (3,146 nodes on the initial mesh), and evaluates Eq. (1) with the sapphire tensor for both modes. The contraction takes 0.44 s. On the initial mesh it matches a central difference of two re-solves of the moved mesh ($\pm 0.5\,\mu$m) to every printed digit: -2.443227×10^{-3} GHz/μm for the qubit and -3.964×10^{-6} GHz/μm for the resonator.

The total_length column. DeviceLayout re-lays the meander when L changes, so no fixed-topology node motion represents it. A transmission-line model holds $f_r(L + \delta)$ constant and takes $\partial f_q/\partial L = 0$, with $\delta = 216\,\mu$m from one finite difference of two geode-fem order-1 solves ($L = 5000$ and $5250\,\mu$m). δ is weakly determined: remesh noise of about 0.15% in either f_r moves it by about $\pm 175\,\mu$m. The model enters only the Jacobian, and every iterate is evaluated on the true objective, so a poorer Jacobian costs steps, not accuracy.

Newton on the high-fidelity objective. DeviceLayout re-meshes every design, so the first-order objective is not smooth in the parameters: from $c = 620$ to $650\,\mu$m the remeshed first-order difference of the qubit is -5.23×10^{-3} GHz/μm against a fixed-topology derivative of -2.28×10^{-3} averaged over the interval (the mean of its end values), because remeshing alone moves the qubit by about 2.6%, more than the 2.0% the step moves it. The iteration therefore runs Newton directly on the high-fidelity objective F, in DeviceLayout's variables (x_1, x_2), in which the frequencies are close to linear and nearly decoupled: a probe in either variable moves its own mode by 4.5% and the other by at most 0.05%. The c column of the Jacobian is geode-fem's first-order derivative scaled by the fidelity ratio $S = F/f_h$ at the same design. Each step costs one geode-fem first-order solve with its gradient (19–25 s) and one high-fidelity evaluation; the geode-fem solves of the one-time δ calibration are outside the Palace-run count.

Against DeviceLayout's discretization. Table 3 and Figure 5(b) compare the optimizers on one objective. Newton with the shape-derivative column meets the criterion at its second step, after 3 Palace runs, and its three geode-fem solves add about 70 s, against about 4 min per Palace run on 32 ranks; a third step reaches err $= 1.4 \times 10^{-7}$ at $(1236.360, 7224.900)\,\mu$m. Two gradient-free

Table 3: Palace runs to DeviceLayout's criterion (err $< 10^{-4}$) at its discretization, all in our Palace environment except the published PRIMA run. Every Palace run a method uses counts, with the initial design as run 1 and probes included, among them the probe that sets the Palace δ (Broyden's second probe, shared by the two rows); geode-fem solves are not counted. Each count is one trajectory. PRIMA runs at `rhobeg` = 0.2 unless stated; the Newton variants use the transmission-line L column. Sources: the `optimize` and `controls` artifacts under `benchmarks/transmon_devicelayout_*/`.

Method	c column	δ from	Runs	err after first step	Design at criterion (μm)
PRIMA, published	none	none	9	unpublished	(1217.678, 7243.955)
PRIMA, our environment	none	none	10	n/a	(1209.125, 7207.551)
`rhobeg` 0.1	none	none	10	n/a	(1235.705, 7216.766)
`rhobeg` $(x_u - x_l)/2$	none	none	12	n/a	(1245.900, 7291.401)
Newton	shape derivative	geode-fem	3	7.9×10^{-4}	(1225.346, 7223.250)
Newton	scaling law	geode-fem	3	1.7×10^{-4}	(1232.664, 7221.270)
Newton	scaling law	Palace	4	2.1×10^{-4}	(1229.026, 7225.712)
Broyden	two probes	none	5	6.2×10^{-4}	(1224.847, 7232.447)

controls test where the 3 runs come from. The first replaces the c column by the scaling law $\partial F_q/\partial x_1 = F_q/x_1$, which follows from $f_q \propto 1/\sqrt{C_\Sigma}$ with C_Σ taken proportional to c. With the geode-fem δ it also takes 3 runs, and its first step has an err 4.7 times lower (Table 3), about 2.2 times closer in frequency. With $\delta = 563.879\,\mu$m taken instead from a Palace probe (c fixed, L from 5000 to 5263 μm), so that no geode-fem input remains, it takes 4 runs: the same two steps, plus the probe. This δ is 2.6 times the geode-fem one, more than the $\pm 175\,\mu$m noise estimate, and costs no extra step. The second control, Broyden's method (Broyden, 1965) from two forward finite-difference probes, takes 5.

PRIMA in our environment. DeviceLayout's own PRIMA call, re-run verbatim in our environment, takes 10: runs 1–5 are its initial interpolation set of $2n + 1 = 5$ points (Powell, 2009), and run 9 misses the criterion by 15% (err $= 1.150 \times 10^{-4}$). In this environment the published final design already meets the criterion (err $= 1.5 \times 10^{-5}$), so the environment moves PRIMA's trajectory, not the attainability of the target; the published count of 9 is from DeviceLayout's environment. With the initial trust radius `rhobeg` at 0.1 instead of 0.2, PRIMA also takes 10 runs, and at the largest radius BOBYQA accepts on DeviceLayout's box, $(x_u - x_l)/2$ (one ulp below 0.4, which PRIMA rejects), it takes 12. Each setting is a single run. In one environment, then, the comparison is 3 Newton runs against 10 to 12, or 4 with no geode-fem input.

Where the first steps land. The shape-derivative and Broyden first steps land at $c = 1338.338$ and 1320.522 μm, past about 1230 μm at the criterion, because both follow a local slope, $\partial F_q/\partial x_1 =$ 3.66 and 3.71 GHz, within about 1.5% of each other and 11–12% shallower than the scaling law's $F_q/x_1 = 4.168$ GHz.

Against the extrapolated frequencies. Starting from the published final design, each Newton step uses the extrapolated estimate and the geode-fem derivative at the current design, and each new design is evaluated by a full ladder with the rule of Section 5 (Table 4). The first overshoots the qubit by -1.09%; the second lands at $+0.16\%$ and -0.04% of the targets, with both targets inside the bands. On both ladders geode-fem and Palace agree on the qubit to 2×10^{-4} %. One scaling-law step from the same start, evaluated by the Palace half of the ladder, lands 3.6 and 8.2 μm short of step 2 and at $+0.13\%$ and $+0.12\%$ of the targets: one gradient-free step does what the two shape-derivative steps did.

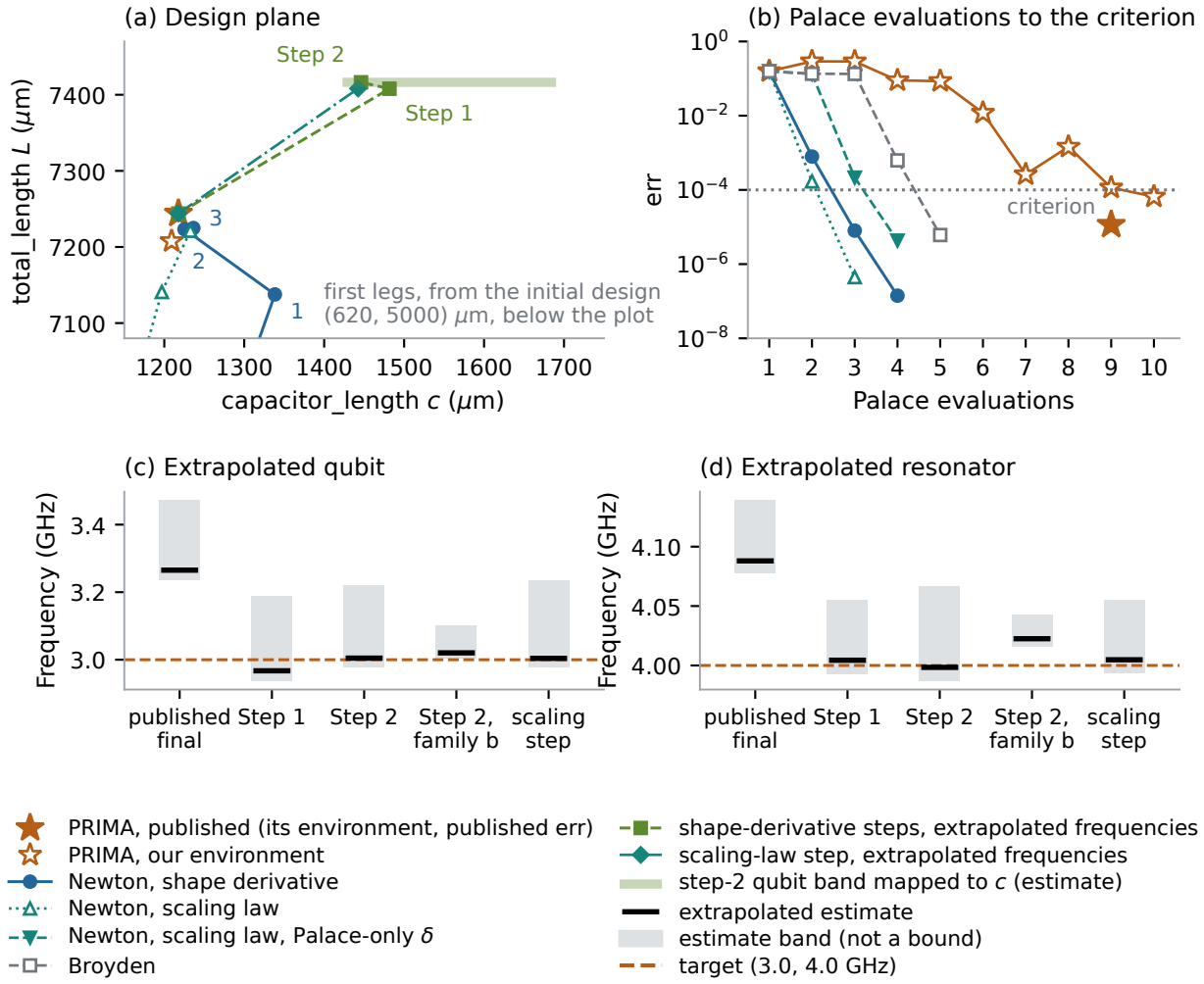


Figure 5: The re-optimization and its controls. (a) The design plane near the optimum: the published final design, PRIMA's end point in our environment, the Newton iterates against DeviceLayout's discretization (numbered: shape derivative; the first legs start off the plot), and the steps against the extrapolated frequencies, with the step-2 qubit band mapped linearly to capacitor_length (an estimate). (b) DeviceLayout's err against Palace evaluations at its discretization in our environment, probes counted (the Palace-δ Newton's run 2 is the L probe it shares with Broyden), PRIMA at `rhobeg` $= 0.2$; the published PRIMA point is at its 9 runs and the err of its published frequencies, 1.151×10^{-5} (DeviceLayout's environment). This counts Palace evaluations, not total cost: each shape-derivative Newton point also costs one geode-fem solve. (c, d) Extrapolated estimates and bands against the targets, with step 2 in both refinement families. Sources: the `optimize`, `controls` and `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/`.

The re-optimized design. Step 2 is the design whose extrapolated frequencies sit on the targets: $c = 1446.078\,\mu$m, 18.8% longer than published, with a resonator 2.4% longer ($+172\,\mu$m). The second refinement family at step 2 gives 3.0203 [3.0033, 3.1014] GHz for the qubit and 4.0225 GHz for the resonator, 0.52% and 0.60% above the first family, the same spread as on the initial geometry; both targets lie just below its band, so on family b step 2 is slightly short. Mapped to 3.0 GHz by the scaling law $f_q \propto c^{-1/2}$, the two families' qubit estimates call for $c = 1451$ and $1466\,\mu$m, +19.1% and +20.4% over the published length. Their bands map to +17% to +37% (family a) and +19% to +27% (family b) under the scaling law, and to +17% to +39% (c from about 1423 to $1690\,\mu$m) and +19% to +28% linearly through $dF/dc \approx S\,\partial f_h/\partial c = -9.05 \times 10^{-4}$ GHz/μm at the step-1 design,

Table 4: Steps against the extrapolated frequencies, in GHz. Each design is evaluated by a full resolution ladder (the scaling-law step and the second family by Palace alone) and the rule of Section 5; values are estimates with their bands. Family b refines step 2's h mesh with Palace's uniform refinement instead of gmsh's. Sources: the `optimize`, `controls` and (published final) `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/`.

Design	c (μm)	L (μm)	f_q [band]	f_r [band]
Published final	1217.678	7243.955	3.265 [3.236, 3.471]	4.088 [4.077, 4.139]
Step 1	1481.657	7407.945	2.967 [2.938, 3.188]	4.004 [3.993, 4.055]
Step 2	1446.078	7416.352	3.005 [2.979, 3.221]	3.998 [3.987, 4.067]
Step 2, family b	1446.078	7416.352	3.020 [3.003, 3.101]	4.022 [4.016, 4.043]
Scaling-law step	1442.479	7408.181	3.004 [2.977, 3.235]	4.005 [3.994, 4.055]

centred on the target. Under either mapping the first range contains the second. Since the estimates lean low, the common-limit fit, which puts step 2 at 3.039 and 4.019 GHz, would call for a longer capacitor still.

8 How geode-fem Was Designed and Built

geode-fem implements the whiteroom operator specification of Crutcher Dunnavant (Dunnavant, 2026), whose four layers, L1–L4, restate Palace's cited source as pure functions, unfold those into compositions of base algebraic primitives, recast them where possible as global tensor operations, and express them in a small graph-evaluation calculus. He also gave design input on the solver's operator layer and contributed code: the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate. The rest of the code, the experiments and the text of this paper were produced by AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by Squad, a message room the agents share. Nearly every change was reviewed by a separate agent session. The agent reviews of this paper recomputed its numbers from the archived artifacts, reproduced runs, and mutation-tested the scripts that check the text and the figures against those artifacts. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

9 Discussion

What is established, and how. Table 5 sorts the results by the kind of evidence behind them. The brackets are the only bounds, and the cap follows from a bracket only under a circuit model.

Why the shape derivative gives no advantage here. The derivative is used at every step of the re-optimization, but three properties of this benchmark remove its advantage. First, the problem has two nearly decoupled parameters, close to linear in DeviceLayout's variables, so Newton's method converges with any reasonable Jacobian: the shape derivative, a two-probe secant, or the scaling law $f_q \propto x_1$, which costs nothing. Second, a finite-difference gradient costs only $N + 1 = 3$ solves. Third, the coarse-mesh, fixed-topology derivative is the slope of a different discrete problem from the one the iteration evaluates (DeviceLayout remeshes every design and Palace evaluates it at second order); once scaled by the fidelity ratio S it gives about what Broyden's probe gives (Section 7). That the derivative is exact for its own discrete problem and cheap (0.44 s) does not change the number of Palace runs.

Table 5: Evidence tiers. Bounded: a two-sided bound that holds by theorem for the meshed polygonal geometry. Model-dependent: a rigorous consequence of a bound under a stated circuit model. Measured: a solve on a stated discretization. Estimated: an extrapolation with a stated rule. Consistency: an agreement that is not independent evidence.

Tier	Results
Bounded	Island capacitance C_{isl} of the unmodified device, without C_J, natural box: $[77.40, 77.61]$ fF (scalar), $[79.56, 79.76]$ fF (tensor); grounded box, the eigenproblem's model: $[79.56, 79.77]$ fF (tensor), so $C_\Sigma = C_{\text{isl}} + C_J$ in $[85.06, 85.27]$ fF and $e^2/2hC_\Sigma$ in $[0.2271, 0.2278]$ GHz; grounded minus natural in $[0, 0.21]$ fF; tensor-minus-scalar gap 2.5–3.1%
Model-dependent	The initial-qubit cap of $4.477\,\text{GHz} = 1/(2\pi\sqrt{L_J C_\Sigma})$ at the lower bound on C_Σ: two-node circuit, quasi-static C_Σ, bounded on the fixture's polygonal geometry, not on the ladder meshes; with the adaptive level, $[4.4297, 4.477]$ GHz; the same $e^2/2hC_\Sigma$ read as the transmon's charging energy E_C/h
Measured	Same-mesh agreement (Table 1); the reproduction within 0.7%; every ladder and adaptive level (adaptive qubits +7.0% and +8.0%, still rising); the gradient check; Palace runs to DeviceLayout's criterion, one trajectory each (Table 3)
Estimated	Extrapolated frequencies and bands (Table 2); the adaptive and common-limit fits; the static-route qubit range; L_s and k; the box-model effect on C_{isl}, 8×10^{-5}; the re-optimized capacitor, +19 to +20% over two families and about +17 to +39% within the band
Consistency	The two routes agree to 0.22% on the initial qubit; the extrapolation rule was fixed after the static route was known, so this is not a blind prediction
Not claimed	Agreement with measurement; a verified hit of 3.0/4.0 GHz in the $h \to 0$ limit; a global optimum; an advantage of the shape derivative on this problem; eigenmode-participation (EPR) derivatives, which need eigenvector derivatives (Nelson, 1976); a speed advantage over Palace

Derivatives pay off in other regimes. With N parameters, a discrete adjoint gives all N derivatives of one objective for about one extra solve, where finite differences need N more; coupled targets with no per-parameter scaling law need a computed Jacobian; and freeform shapes have many parameters by construction. A companion paper on adjoint shape optimization of a conformal patch antenna (Walters, 2026) measures this: at 38 shape parameters the adjoint needs at least 11 times fewer solves than the best gradient-free run, tuned settings included, to reach its -9.5 dB target, the ratio falls to 1.1–2.4 at 4 parameters, and a tuned full-covariance CMA-ES comes within a factor of 7.7 at -9 dB.

Threats and limitations. Agreement between two codes is evidence against implementation error, not against shared modelling choices such as the port sheet behind the 17.49 GHz mode. The extrapolations assume the rate the sheet-edge singularity allows on uniform meshes, their high sides are wide, neither adaptive run reached a plateau, and L_s and k were extrapolated from two or three discretizations. Every limit $h \to 0$ here is a limit of DeviceLayout's model (zero-thickness PEC sheets, lossless and linear, open readout ports in the ladders); film thickness and kinetic inductance remove the edge singularity and move the frequencies by amounts not measured here. The element edges along the sheet outlines are mostly 10–12 μm long on h and so 2.5–3 μm on $h/4$, more than ten times a 0.2 μm film: every mesh used here sees the zero-thickness edge, and the shift measured here is the discretization error within the model, for one device.

What the shift means for a design. Table 6 reads the initial qubit as a total capacitance. The bound gives $E_C/h = e^2/2hC_\Sigma$ in $[0.2271, 0.2278]$ GHz for the grounded box, Bounded as a function of C_Σ and model-dependent as the transmon's charging energy (a single-island lumped picture). Under the two-node circuit with the L_s and k bands, the published 4.14 GHz implies C_Σ

in $[98.59, 98.72]$ fF, 15.6–16.1% above the bound, so E_C and the anharmonicity, close to $-E_C$ (Koch et al., 2007), are larger than the published discretization implies by as much. In the bare LC reading at fixed L_J ($L_s = k = 0$), which gives a larger C_Σ than the two-node one for every frequency, the published 99.45 fF exceeds the bound by 16.6–17.0%, the static-route reading by 15.5–15.9% and the adaptive-level reading by 14.5%. Of that factor of 1.166–1.169, the measured adaptive level accounts for 1.145, the gap between that still-rising level and the static route for 0.9–1.3% more, and L_s and k, which the bare reading omits, for the remaining 0.6–1.2%. The transition f_{01} rises by a little less than the linear mode, about 6.6% at the adaptive level. L_J can retune the qubit frequency but not E_C, which the capacitance alone sets.

Table 6: The initial qubit read as a total capacitance C_Σ, with $E_C/h = e^2/2hC_\Sigma$; "bare" is the LC reading at fixed L_J ($L_s = k = 0$), "two-node" the circuit of Section 3.3 with the L_s and k bands, given first for the published value. The bound is for the grounded box. Sources: `c_sigma_bracket_grounded_box.toml`, `benchmarks/transmon_loop_inductance/results.toml` and the adaptive run of Table 2.

Source	Tier	f_q (GHz)	C_Σ (fF)	E_C/h (GHz)
Bound	Bounded	–	$[85.06, 85.27]$	$[0.2271, 0.2278]$
Static route	Estimated	$[4.449, 4.458]$	85.8–86.1 (bare)	0.225–0.226 (bare)
Adaptive finest level	Measured	4.4297	86.87 (bare)	0.223 (bare)
Published	–	4.14	$[98.59, 98.72]$; 99.45 (bare)	0.196; 0.195 (bare)

What a practitioner can do. The slow rate comes from the sheet edges, and the standard remedy for an edge singularity is to refine toward it, with a mesh graded toward the edges or with *hp* refinement (Demkowicz, 2006), which recover faster convergence than uniform refinement. Palace's adaptive runs of Section 5 are measured instances, and the gap-refined, fourth-order setting of Ye et al. (2025), within 0.3% of their extrapolated values (Section 2), is a published one. A first check is the strip width: on the fixture, 993 of the 1,007 edges that trigger Palace's bisection lie where a resonator or feedline strip is one element wide (Section 4). The same edge fields set the surface participation that limits coherence, computed with a two-step finite-element simulation (Wang et al., 2015) or analytical formulas for finite metal thickness (Martinis, 2022); such loss estimates need the same edge resolution as the frequencies.

10 Conclusion

Palace reproduces the published DeviceLayout.jl SingleTransmon frequencies at the example's own discretization, which sits 6.2–8.4% low on the qubits: adaptive runs put the two qubits 7.0% and 8.0% above their published values with no extrapolation, and the estimates of two refinement families, which lean low, put them 7.3–9.6% above. geode-fem reproduces Palace's qubit on every level both codes ran, and a two-sided bracket on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model. At the extrapolated frequencies the 3.0/4.0 GHz design needs a capacitor about 19–20% longer than published, or 17–39% within the band. Newton's method reaches the example's criterion in 3 Palace runs with geode-fem input (its shape derivative, or its calibration of the resonator model) and in 4 without, against 10 to 12 for PRIMA. For design examples that are meant to be copied, the discretization is part of the result and belongs in the report beside it.

11 Reproducibility

Every result is read from an archived, scripted artifact.[1] Each caption names its artifact; the circuit terms L_s and k come from `benchmarks/transmon_loop_inductance/` and `benchmarks/transmon_coupling/`, the Palace cells are under `reference/fixtures/transmon_palace/`, and the DeviceLayout meshes are generated by `reference/julia/generate_transmon_param.jl`. The versions are geode-fem commit `3174015` for the same-mesh agreement, with the commit of each later result recorded in its artifact, Palace `fba6a5b`, DeviceLayout.jl v1.15.0, PRIMA.jl 0.2.4 and Burn 0.21. Palace ran on AWS Graviton4 hosts (32 and 16 vCPU; the final-design adaptive run on 24 ranks); the capacitance brackets ran on a 28-core, 96 GiB workstation and their grounded-box re-solves on a Graviton4 host. geode-fem's direct solves at $h/2$ peaked at 94 and 136 GB, so only Palace ran the larger cells. Palace's ladder cells use SLEPc shift-invert at tolerance 10^{-8}, `Target` 3.0 (initial) or 2.0 GHz (final), and its default direct solver, SuperLU (order-1 h), or AMS-preconditioned GMRES at relative tolerance 10^{-9}, on 16 (order-1 h) or 32 MPI ranks; the fixture ran on 8, and the runs at DeviceLayout's discretization keep the example's tolerances (10^{-6} and 10^{-7}, `Target` 2.5 GHz). geode-fem's Lanczos shifts are 5 and 4 GHz, with relative residuals below 4×10^{-8}. The h, $h/2$ and $h/4$ meshes have 133,355, 1,066,840 and 8,534,720 tetrahedra (initial) and 163,431, 1,307,448 and 10,459,584 (final). The meshes were made with gmsh 4.15.2 (`gmsh_jll`) under Julia 1.10.11, and Palace was built against the MPICH of the Rocky Linux 9 image in `reference/palace/docker/Dockerfile`.

Acknowledgments. We thank Crutcher Dunnavant for the whiteroom specification, for his design input to the solver's operator layer and for his contributions to the codebase, and the Palace and DeviceLayout.jl developers for publishing the SingleTransmon example and its frequencies. The AI-orchestrated development of geode-fem and of this paper is described in Section 8.

References

Volkan Akçelik, George Biros, Omar Ghattas, David Keyes, Kwok Ko, Lie-Quan Lee, and Esmond G Ng. Adjoint methods for electromagnetic shape optimization of the low-loss cavity for the International Linear Collider. *Journal of Physics: Conference Series*, 16:435–445, 2005. doi: 10.1088/1742-6596/16/1/059.

Robert Anderson, Julian Andrej, Andrew Barker, Jamie Bramwell, Jean-Sylvain Camier, Jakub Cerveny, Veselin Dobrev, Yohann Dudouit, Aaron Fisher, Tzanio Kolev, Will Pazner, Mark Stowell, Vladimir Tomov, Ido Akkerman, Johann Dahm, David Medina, and Stefano Zampini. MFEM: a modular finite element methods library. *Computers & Mathematics with Applications*, 81:42–74, 2021. doi: 10.1016/j.camwa.2020.06.009.

AWS Center for Quantum Computing. DeviceLayout.jl: computer-aided design of quantum integrated circuits. `https://github.com/aws-cqc/DeviceLayout.jl`, 2025a. Software, v1.15.0.

AWS Center for Quantum Computing. DeviceLayout.jl: Single-Transmon Simulation. DeviceLayout.jl v1.15.0 documentation and `examples/SingleTransmon/SingleTransmon.jl`, `https://aws-cqc.github.io/DeviceLayout.jl/stable/examples/singletransmon/`, 2025b.

Daniele Boffi. Finite element approximation of eigenvalue problems. *Acta Numerica*, 19:1–120, 2010. doi: 10.1017/S0962492910000012.

Dietrich Braess and Joachim Schöberl. Equilibrated residual error estimator for edge elements. *Mathematics of Computation*, 77 (262):651–672, 2007. doi: 10.1090/s0025-5718-07-02080-7.

C. G. Broyden. A class of methods for solving nonlinear simultaneous equations. *Mathematics of Computation*, 19(92):577–593, 1965. doi: 10.1090/S0025-5718-1965-0198670-6.

Leszek Demkowicz. *Computing with hp-Adaptive Finite Elements, Volume 1: One and Two Dimensional Elliptic and Maxwell Problems*. Chapman and Hall/CRC, 2006. doi: 10.1201/9781420011685.

Crutcher Dunnavant. palace_whiteroom. `https://github.com/crutcher/palace_whiteroom`, 2026. Software repository. Accessed 2026-10-13.

Axel M. Eriksson, Lukas J. Splitthoff, Harsh Vardhan Upadhyay, Pietro Campana, Niranjan Pittan Narendiran, Kunal Helambe, Linus Andersson, and Simone Gasparinetti. Automated, physics-guided, multi-parameter design optimization for superconducting quantum devices, 2025. URL `https://arxiv.org/abs/2508.18027`.

Alexandre Ern and Martin Vohralík. Polynomial-degree-robust a posteriori estimates in a unified setting for conforming, nonconforming, discontinuous galerkin, and mixed discretizations. *SIAM Journal on Numerical Analysis*, 53(2):1058–1081, 2015. doi: 10.1137/130950100.

[1] The geode-fem source is public at `https://github.com/rjwalters/geode-fem`. Archival DOI to be added at deposit.

Sungjun Eun, Seong Hyeon Park, Kyungsik Seo, Kibum Choi, and Seungyong Hahn. Shape optimization of superconducting transmon qubits for low surface dielectric loss. *Journal of Physics D: Applied Physics*, 56(50):505306, 2023. doi: 10.1088/1361-6463/acf7cf.

R. L. Fox and M. P. Kapoor. Rates of change of eigenvalues and eigenvectors. *AIAA Journal*, 6(12):2426–2429, 1968. doi: 10.2514/3.5008.

Sarah Quiñones El Kazdadi. faer: A linear algebra library for the Rust programming language. *Journal of Open Source Software*, 11(123):6099, 2026. doi: 10.21105/joss.06099.

Jens Koch, Terri M. Yu, Jay Gambetta, A. A. Houck, D. I. Schuster, J. Majer, Alexandre Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Charge-insensitive qubit design derived from the Cooper pair box. *Physical Review A*, 76(4):042319, 2007. doi: 10.1103/PhysRevA.76.042319.

Richard B. Lehoucq, Danny C. Sorensen, and Chao Yang. *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*. SIAM, 1998. doi: 10.1137/1.9780898719628.

John M. Martinis. Surface loss calculations and design of a superconducting transmon qubit with tapered wiring. *npj Quantum Information*, 8:26, 2022. doi: 10.1038/s41534-022-00530-6.

Josef Meixner. The behavior of electromagnetic fields at edges. *IEEE Transactions on Antennas and Propagation*, 20(4):442–446, 1972. doi: 10.1109/TAP.1972.1140243.

Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, and Michel H. Devoret. Energy-participation quantization of Josephson circuits. *npj Quantum Information*, 7:131, 2021a. doi: 10.1038/s41534-021-00461-8.

Zlatko K. Minev, Thomas G. McConkey, Maika Takita, Antonio D. Córcoles, and Jay M. Gambetta. Circuit quantum electrodynamics (cQED) with modular quasi-lumped models, 2021b. URL https://arxiv.org/abs/2103.10344.

Jean-Claude Nédélec. Mixed finite elements in $\mathbb{R}^3$. *Numerische Mathematik*, 35(3):315–341, 1980. doi: 10.1007/BF01396415.

Richard B. Nelson. Simplified calculation of eigenvector derivatives. *AIAA Journal*, 14(9):1201–1205, 1976. doi: 10.2514/3.7211.

N.K. Nikolova, J.W. Bandler, and M.H. Bakr. Adjoint techniques for sensitivity analysis in high-frequency structure CAD. *IEEE Transactions on Microwave Theory and Techniques*, 52(1):403–419, 2004. doi: 10.1109/tmtt.2003.820905.

William L. Oberkampf and Christopher J. Roy. *Verification and Validation in Scientific Computing*. Cambridge University Press, 2010. doi: 10.1017/cbo9780511760396.

Palace developers, AWS Center for Quantum Computing. Palace: PArallel LArge-scale Computational Electromagnetics. https://github.com/awslabs/palace, 2023. Software, commit fba6a5b.

Greg Peairs and Hugh Carson. Design quantum integrated circuits with open-source software DeviceLayout.jl from AWS. https://aws.amazon.com/blogs/quantum-computing/design-quantum-integrated-circuits-with-open-source-software-devicelayout-jl-from-aws/, 2025. AWS Quantum Technologies Blog.

M. J. D. Powell. The BOBYQA algorithm for bound constrained optimization without derivatives. Technical Report DAMTP 2009/NA06, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, 2009.

W. Prager and J. L. Synge. Approximations in elasticity based on the concept of function space. *Quarterly of Applied Mathematics*, 5(3):241–269, 1947. doi: 10.1090/qam/25902.

Taha Rajabzadeh, Zhaoyou Wang, Nathan Lee, Takuma Makihara, Yudan Guo, and Amir H. Safavi-Naeini. Analysis of arbitrary superconducting quantum circuits accompanied by a Python package: SQcircuit. *Quantum*, 7:1118, 2023. doi: 10.22331/q-2023-09-25-1118.

P. A. Raviart and J. M. Thomas. A mixed finite element method for 2-nd order elliptic problems. In *Mathematical Aspects of Finite Element Methods*, volume 606 of *Lecture Notes in Mathematics*, pages 292–315. Springer, 1977. doi: 10.1007/bfb0064470.

Lewis Fry Richardson. The approximate arithmetical solution by finite differences of physical problems involving differential equations, with an application to the stresses in a masonry dam. *Philosophical Transactions of the Royal Society of London. Series A*, 210:307–357, 1911. doi: 10.1098/rsta.1911.0009.

Patrick J. Roache. Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116 (3):405–413, 1994. doi: 10.1115/1.2910291.

Sadman Shanto, Andre Kuo, Clark Miyamoto, Haimeng Zhang, Vivek Maurya, Evangelos Vlachos, Malida Hecht, Chung Wa Shum, and Eli Levenson-Falk. SQuADDS: A validated design database and simulation workflow for superconducting qubit design. *Quantum*, 8:1465, 2024. doi: 10.22331/q-2024-09-09-1465.

David Sommers, Zach Degnan, Divita Gautam, Yi-Hsun Chen, Chun-Ching Chiu, Arkady Fedorov, and Prasanna Pakkiam. Open-source highly parallel electromagnetic simulations for superconducting circuits, 2025. URL https://arxiv.org/abs/2511.01220.

Ruben Specogna. Complementary geometric formulations for electrostatics. *International Journal for Numerical Methods in Engineering*, 86(8):1041–1068, 2011. doi: 10.1002/nme.3089.

Tracel AI. Burn: a comprehensive deep-learning and tensor-computation framework in Rust. https://github.com/tracel-ai/burn, 2026. Software, v0.21.0.

Robb Walters. Loom: AI-powered development orchestration. https://github.com/rjwalters/loom, 2025. Software repository. Accessed 2026-10-13.

Robb Walters. Adjoint shape optimization of a conformal patch antenna, 2026. Working title. Preprint in preparation, to be deposited on Zenodo.

C. Wang, C. Axline, Y. Y. Gao, T. Brecht, Y. Chu, L. Frunzio, M. H. Devoret, and R. J. Schoelkopf. Surface participation and dielectric loss in superconducting qubits. *Applied Physics Letters*, 107(16):162601, 2015. doi: 10.1063/1.4934486.

Ziang Wang, Feng Wu, Hui-Hai Zhao, Xin Wan, and Xiaotong Ni. SuperGrad: a differentiable simulator for superconducting processors. *Quantum*, 9:1722, 2025. doi: 10.22331/q-2025-04-24-1722.

Jiale Ye, Jiaheng Wang, and Yu-xi Liu. Electromagnetic feature extraction in superconducting quantum circuits: An open-source finite-element workflow using Palace, 2025. URL https://arxiv.org/abs/2511.09041.

Zaikun Zhang. PRIMA: Reference implementation for Powell's methods with modernization and amelioration. http://www.libprima.net, 2023. Software.